

Data Integration with ETL

This training covers the basics of the CCH® Tagetik ETL module configuration and provides best practices. The attendees will configure an ETL with mapping functionality that can load financial data for a Tagetik implementation. With this knowledge, you can also update existing ETL's with new functionality according to changes and/or new requirements.

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EXERCISE 1 - Setup Staging Table and Parameters

Objective

All activities in ETL use the Staging Table. The Staging table therefore can be considered to be the core of the ETL. For this reason, here we learn how to define a Staging Table. To save time, only 3 columns of the staging table will be defined.

Tasks

Perform the following tasks:

1. Login with username and password:
   - Username: **Yoda**
   - Password: **tgk**
2. On the Admin homepage, click the **ETL** tile and next the Domains tile.
3. Select domain **0_ETL_TRAINING** and click **Dimensions and Dictionaries** from the definitions section in the right hand part of the window.
4. Add 3 more records, according to the list below.
   1. Record 1:
      1. **Code**: 51
      2. **Description**: T_Scenario
      3. **Acronym**: T_Sce
      4. **Dimension Type**: String
      5. **Identity**: Scenario
   2. Record 2:
      1. **Code**: 52
      2. **Description**: T_Period
      3. **Acronym**: T_Per
      4. **Dimension** Type: String
      5. **Identity**: Period
   3. Record 3:
      1. **Code**: 53
      2. **Description**: T_Entity
      3. **Acronym**: T_Ent
      4. **Dimension Type**: String
      5. **Identity**: None

Then, perform the following tasks:

1. Save the changes and close the window.



2. Select domain0_ETL_TRAININGand click **Parameters** from the definitionframein the right hand part of the window.
3. Create 3 records, all with **Type: Imported Data** , according to the list below.
    1. Record 1:
        1. **Code**: PAR_TGK_SCE
        2. **Description**: Parameter Tagetik Scenario
        3. **Dimension**: 51 - T_Scenario
        4. **Multiselection**: No
    2. Record 2:
        1. **Code**: PAR_TGK_PER
        2. **Description**: Parameter Tagetik Period
        3. **Dimension**: 52 - T_Period
        4. **Multiselection**: Yes
    3. Record 3:
        1. **Code**: PAR_TGK_ENT
        2. **Description**: Parameter Tagetik Entity
        3. **Dimension**: 53 - T_Entity
        4. **Multiselection**: Yes

Create an Imported Data Intersection Parameter:

- Code: **PAR_TGK_SCE_PER**
- Description: **Parameter Tagetik Scenario/Period**
- Multiselection: **Ticked**

Access parameter **PAR_TGK_PER** and **PAR_TGK_SCE** and for each one set **PAR_TGK_SCE_PER** as Parent to have a Scenario/Period parameter

EXERCISE 2 - Setup Extraction

Objective	Learn how to setup an extraction for an Excel source.
Tasks	Perform the following tasks:

1. On the Admin homepage, click the **ETL** tile and next the Domains tile.
2. Select domain **0_ETL_TRAINING** and click **Data sources** from the definitionsframein the right hand part of the window.
3. Create a record with the following details within the Attributes tab:
   - Code: **010**
   - Description: **Extract Data from Excel**
   - Type: **Excel (xlsx)**
   - Area: **DATA**
4. Open the **Fields definition** of the new created datasource
5. Add new records following the list below

Dimension	Name of the Field on Excel File (xslx)
1	Year
2	Period
3	Entity
4	Account
5	Costcentre
6	Product
10	Amount
21	Version
22	Source

Need any help? Here are some hints

- To assign the Excel type to the record created, switch to the **Files** tab and put a mark in the checkbox in front of Excel (xlsx).
- To set the Area within the Excel file, enter **DATA** in the **Excel area (xlsx)** field and click the 'Save' button.

EXERCISE 3 - Lookup Tables

Objective

CCH Tagetik offers multiple ways to modify (Transform) the extracted data to match the Tagetik data model.

One of them is the use of Lookup or Mapping tables.

Tasks

Perform the following tasks:

1. On the Admin homepage, click the **ETL** tile and next the Domains tile.
2. Select domain **0_ETL_TRAINING** and click **Lookup tables** from the definitions section in the right hand part of the window.
3. Add a new record with the following details:
   - Code: **040**
   - Description: **Account Mapping**
   - Input dimension 1: **4 - S_Account**
   - Output dimension 1: **54 - T_Account**
   - Output dimension 2: **31 - TMP_Acc_SignChange**
4. Open the new created record and add new records according to the following table:

Input	Output 1	Output 2
AA001	10010	1
DA001	50010	-1
XA001	90210	1
XA004	90240	1

5. Save and Close.

Need any help? Here are some hints

- To reach the Lookup Tables, select domain0_ETL_ TRAININGand click **Lookup tables** from the definitions section in the right hand part of the ETL Domains window.
- To reach the window where to add the new records, first Click the **Elements** link in the Details section.



EXERCISE 4 - Setup Loading

Objective

Data can be loaded into several tables of CCH Tagetik. In this exercise, we will load data in the Original Amounts table: DATI_SALDI_LORDI.

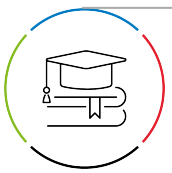
Tasks

Perform the following tasks:

1. On the Admin homepage, click the **ETL** tile and next the Domains tile.
2. Select domain **0_ETL_TRAINING** and click **Data target: CCH Tagetik** from the definitions section in the right hand part of the window.
3. Create a new record with the following details:
   - Code: **010**
   - Description: **Load Gross Amounts Tagetik**
   - Table: **DATI_SALDI_LORDI - Gross amounts**
4. Open the Fields definition of the new created loading
5. Add new records according to the following table:

Field	Update Type	Dimension/Value	Group By	Aggregate Function
COD_AZIENDA	By Dimension	53	Yes	
COD_CATEGORIA	By Dimension	50	Yes	
COD_CONTO	By Dimension	54	Yes	
COD_DEST1	By Dimension	55	Yes	
COD_DEST2	By Dimension	56	Yes	
COD_PERIODO	By Dimension	52	Yes	
COD_SCENARIO	By Dimension	51	Yes	
COD_VALUTA	By Dimension	61	Yes	
COD_VALUTA_ORIGINARIA	By Dimension	61	Yes	
IMPORTO	By Dimension	60		SUM
IMPORTO_VALUTA_ORIGINARIA	By Dimension	60		SUM

6. Save and Close.



EXERCISE 5 - Create ETL Routine

Objective

An ETL Routine links all actions of ETL together. It is a series of steps to do in a given order to Extract, Transform and Load the data. Here we learn how to create an ETL Routine.

Tasks

Perform the following tasks:

1. On the Admin homepage, click the **ETL** tile and next the Domains tile.
2. Select domain **0_ETL_TRAINING** and click **Routines** from the definitions section in the right hand part of the window.
3. Create a new routine with the following details:
   - Routine: **110**
   - Description: **Import Gross Amounts**
   - Process: **check**
   - Store Transformation: **check**
   - Store Extraction: **check**
   - Warning management: **Show**
   - Leave all other options at their default values.
4. Save this routine.

Then, perform the following tasks:

1. Open Extraction from the Definition section on the Attributes tab and add an extraction
   - Sorting: **10**
   - Data source: **010 Extract Data From Excel file**
   - Description: **Extract Data from Excel**
2. Save the extraction and close the Extractions window
3. Open Transformations from the Definition section on the Attributes tab.
4. Add a new transformation
   - Sorting: **10**
   - Task: **Enrichment SQL Instruction**
   - Description: **Set Tagetik Scenario**
   - Output dimension: **51 – T_Scenario**
   - Input text (specific): **concat(CAMPO1, 'ACT')**
5. Add a new transformation
   - Sorting: **20**
   - Task: **Enrichment Lookup Table**
   - Description: **Set Tagetik Account and Sign change**
   - Lookup table: **040 – Account Mapping**
6. Add a new transformation

- Sorting: **30**
- Task: **Enrichment SQL Instruction**
- Description: **Set Tagetik Period**
- Output dimension: **52 - T_Period**
- Input text (Specific): **CAMPO2**

7. Add a new transformation
    - Sorting: **40**
    - Task: **Enrichment SQL Instruction**
    - Description: **Set Tagetik Costcentre**
    - Output dimension: **55 - T_Costcentre**
    - Input text (specific): **CAMPO5**
8. Add a new transformation
    - Sorting: **50**
    - Task: **Enrichment SQL Instruction**
    - Description: **Set Tagetik Product**
    - Output dimension: **56 - T_Product**
    - Input text (specific): **CAMPO6**
9. Add a new transformation
    - Sorting: **60**
    - Task: **Enrichment SQL Instruction**
    - Description: **Set Tagetik Category**
    - Output: **50 - T_Category**
    - Input text (specific): **'$AMOUNT'**
10. Add a new transformation
    - Sorting: **70**
    - Task: **Enrichment SQL Instruction**
    - Description: **Set Tagetik Entity**
    - Output: **53 - T_Entity**
    - Input text (specific): **CAMPO3**
11. Add a new transformation
    - Sorting: **80**
    - Task: **Enrichment SQL Instruction**
    - Description: **Set Tagetik Currency**
    - Output: **61 - T_Currency**
    - Input test (specific):

select COD_VALUTA from AZIENDA where COD_AZIENDA = CAMPO53

12. Add a new transformation
    - Sorting: **90**
    - Task: **Enrichment SQL instruction**
    - Description: **Set Tagetik amount and apply sign change**
    - Output dimension: **60- T_Amount**
    - Input text (specific): **CAST((CAST(CAMPO10 AS FLOAT) * CAST(CAMPO31 AS FLOAT)) AS VARCHAR (50))**
13. Save the transformations and close the Transformations window
14. Open Loadings from the Definition section on the Attributes tab and add a loading.
    - Sorting: **10**
    - Data Target: **CCH Tagetik: 010 Load Gross Amounts Tagetik**
    - Description: **Load Gross Amounts Tagetik**
    - Loading Policy: **Update Data (delete/insert)**
15. Set the following parameters on the 'Filter parameters' tab:
    - Parameter 1: **PAR_TGK_SCE**
    - Parameter 2: **PAR_TGK_PER**
    - Parameter 3: **PAR_TGK_ENT**
16. Set the following ETL Data Filter on the 'Filters' tab:
    - **CAMPO60 IS NOT NULL AND CAMPO60 <> '0'**
17. Save the loading and close the Loadings window

Need any help? Here are some hints

- To create a new Extraction rule, open **Extractions** from the Definition section on the Attributes tab of the routine.
- To create a new Transformation rule, open **Transformations** from the Definition section on the Attributes tab of the routine.
- To create a new Loadings rule, open **Loadings** from the Definition section on the Attributes tab of the routine.
- In the file Transformations.txt that you can find in the folder Workshops\ETL material it is possible



copy/paste the input text of Transformations "Set Tagetik currency" and "Set Tagetik amount and apply sign change"

EXERCISE 6 - Create ETL Job

Objective

An ETL job is a collection of routines. Running a job requires less user input compared to running a routine.

Here we learn how to create an ETL job.

Tasks

Perform the following tasks:

1. On the Admin homepage, click the ETL tile and next the Jobs tile.
2. Add a new record with the following settings
   - Code: **700**
   - Description: **ETL Data**
3. Save the record
4. Click **Routines** in the details section
5. Add a new record in the details window with the following settings
   - Sorting: **10**
   - Enable: **Yes**
   - ETL Domain: **0_ETL_TRAINING**
   - Routine: **010**
6. Save the record and close the Routines window

Then, perform the following tasks:

1. Click **Parameters** in the details section
2. Change all parameters to have 'Define automatically' enabled.
3. Save the changes and close the Parameters window
4. Click **Extractions** in the details section
5. Choose Update extractions from the action menu button
6. Set the Data source type to **Excel (xlsx)**
7. Set the Extraction type to **Client**
8. Save the changes and close the Extractions window



EXERCISE 7 - Job Execution

Objective

Execute an ETL job.

Tasks

Perform the following tasks:

1. In Excel, open **ETL_Source_File.xlsx** from the folder C:\WORKSHOPS\Data integration with ETL of your training clone.
2. Select all cells with content (not the entire worksheet)
3. Assign the name **DATA** to this selection
4. Save the Excel file on the **Desktop**.

After that, perform the following tasks:

1. In CCH Tagetik, Go to the User Homepage.
2. Click the **ETL tile** and after that the **Execution** tile
3. In the appearing window, select Process: **2050ACT - 2050 Actual** and Select Job: **700 - ETL Data**
4. In the bottom-left corner, the Uploads tab is automatically selected
5. Click the **Load files** button and select the previously prepared Excel file, then click **Continue**.
6. Click **Run**
7. Data processing is started. Wait until finished and the result window appears.
8. Check the warnings by going to the Detail tab and expand the warning until the details are shown (Warnings are identified through the ⚠ icon).

Need some help?



Here you can find the solutions to the most frequent technical issues you may get stuck with during the course.

I can't login to the training environment

Make sure you are using the access link provided by the trainer (with the proper 4 digits token if needed!). Events have a starting and ending date/time.

Make sure you are not trying to access the training course too early or too late!

My machine is slow

- Keep in mind that lab configuration may require up to 20 minutes to start from the moment you click on the CCH Tagetik Chrome icon! 

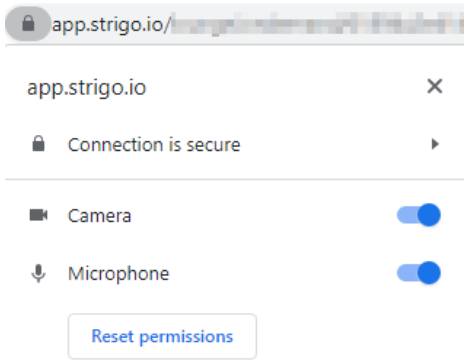
- The local connection's bandwidth is not sufficient for the number of connections currently in place (performance issues on the server side are very rare).

  Solution: ask the IT services to reboot the access point and prefer connection via cable if available.

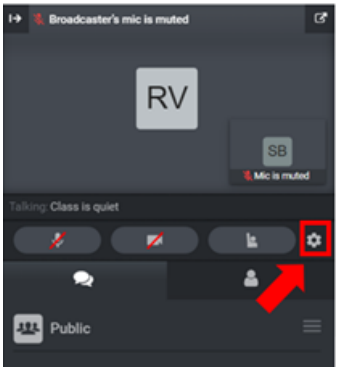
Audio/Video Troubleshooting

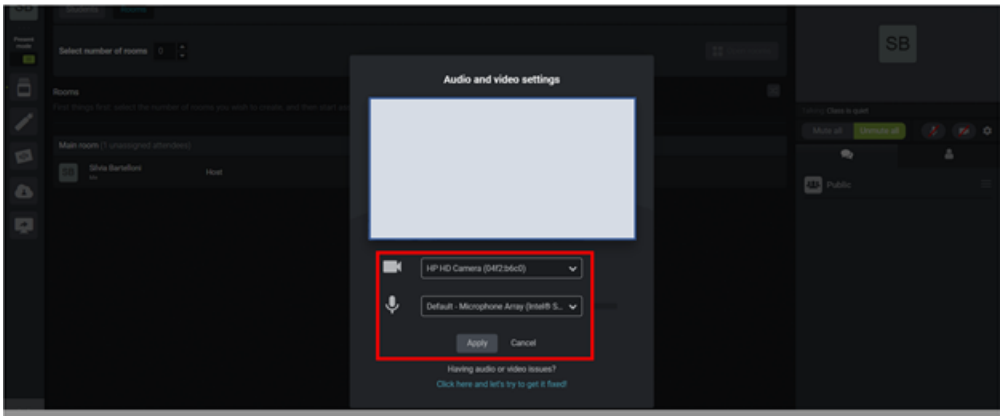
In case you are having problems with audio and video try the following:

1. Make sure you are running a compatible browser (Google Chrome, Mozilla Firefox).
2. Make sure your media devices are connected and are not muted or set to very low volume.
3. Make sure the browser you are in is allowed to use your laptop audio and video



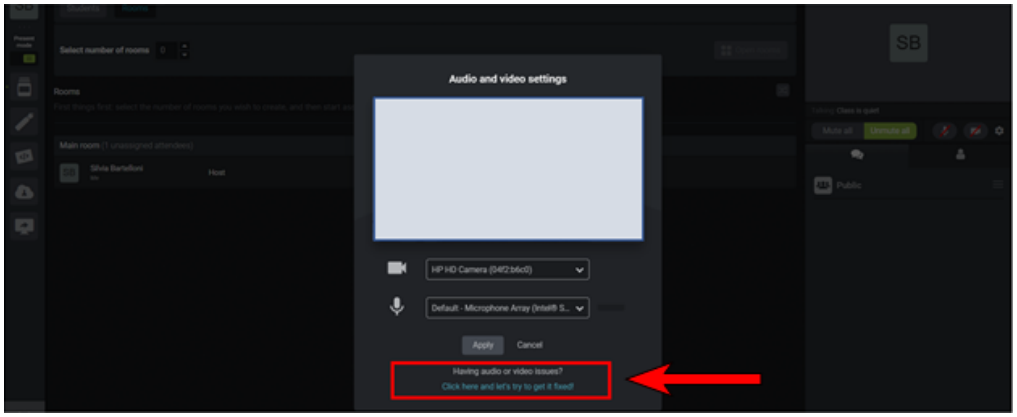
4. Make sure no other applications are running that use your media devices (microphone, camera or speakers).
5. Prevent any ad-blockers or script blockers running on the classroom's tab.
6. Refresh the page.
7. From the right communication panel, click on the settings icon and change the camera/video options selected





Check if there are any errors: if so, they will appear in red.

In addition, you can run a troubleshoot by clicking on: "Let's try and get it fixed" and then on "Run Troubleshoot"



If you still have problems, it could depend on network restrictions.

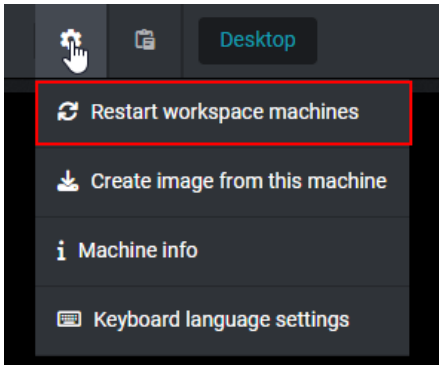
Resolution issues: if you see CCH Tagetik application masks too big in the machine even if I didn't change browser zoom.
Remember, Strigo adapts the machine to your computer resolution. Try to change your resolution or use a different monitor!

⚠ **Important:** Scaling setting recommended = 100% (max 125%).

Lab troubleshooting while practicing

If you are in your Virtual Machine and you are experiencing a **black screen** with your lab not starting:

1. Try restarting the machine by clicking on the wheel icon and then on "Restart workspace machine"



2. Try logging out and then log in again after few minutes

⚠ **Important:**

Your machine will be rebooted, but do not panic! No data will be lost. You will need to wait for the OS to restart and CCH® Tagetik to be reloaded though.

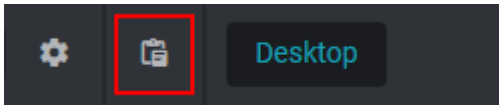
3. A pause > restart of the lab should work as well. The pause option is on the top right-hand corner of the screen. The workspace will switch to **Paused** mode, and then you want to really just give it around two minutes. This is usually enough time to let the lab machine be actually stopped, before you go and hit **Resume** again.

4. Try to open the browser in incognito/private mode.
5. Reach out to Strigo support through this link and then click start live chat

I can't copy/paste any content to the virtual machine

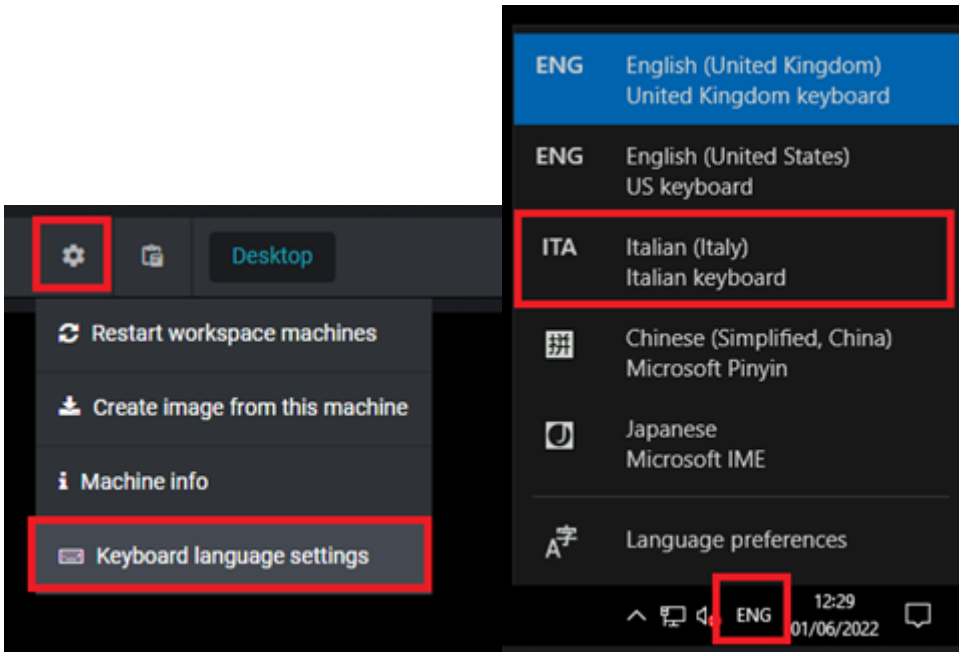
Direct copy/paste is not allowed.

Solution: Use the clipboard icon to copy and paste from the desktop to the virtual machine and viceversa.



I cannot change language in my machine

Solution: Change the language both by desktop bar and by Settings.



MS Excel: I can't select a given form or an object of a form.

This issue happens pretty often and is due to some wizard (e.g. Matrix Wizard, Cell Fields wizard) already open but not currently diplayed on the screen. When any wizard is open and running, you can't work anywhere else.

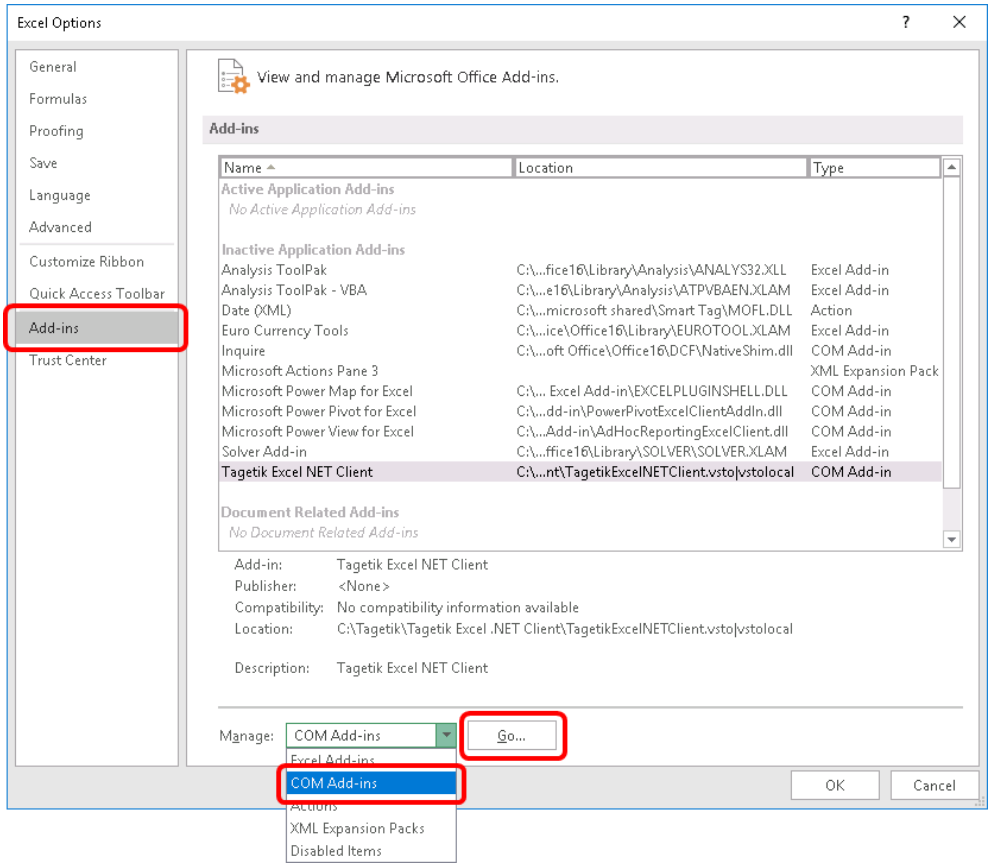
Solution: check if you have any of the wizard open and running (most common: Matrix Wizard, Tab Filters, Form Filters, Cell Fields wizards). If this is the case, be sure to have finished your work on that wizard and close it with OK to confirm. Then go on with your work.

MS Excel: the Tagetik menu is no longer available on the ribbon bar.

Solution1: this probably means the CCH Tagetik add-in has been disabled by MS Excel due to a crash or some errors and has to be enabled again.

Follow the path **File > Options**. Click "**Add-Ins**" on the "**Excel Options**" pane and expand the drop-down menu "**Manage**" on the bottom. Try to select:

1. "**COM Add-Ins**" > "**Go...**". If unticked, tick the "**Tagetik Excel NET Client**" Add-in and click OK
2. "**Disabled items**" > "**Go...**". If you find there the "**Tagetik Excel NET Client**" Add-in, enable it. Close and start Excel.



Solution2: should solution1 not solve the issue, disinstall and reinstall the CCH® Tagetik add-in. Ask the trainer for some help in case.

If you use a 4K resolution monitor, there's a chance you will see far too little

Solution: Low the resolution before the training, then set it back to the original value after the training

My problem is not listed here.



If even this can't solve your problem, please refer to your trainer (and if you are a trainer to your training manager).